

Literature Review

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Generative artificial intelligence has changed the way many organizations think about productivity and information management. McKinsey & Company (2023) explains that generative AI has expanded quickly because it can support tasks such as writing, summarizing, analyzing information, and improving workflow efficiency. This is important for professional industries because many employees now have the access to AI tools that can help with daily tasks without requiring advanced technical knowledge. Instead of only being used for highly technical applications, AI is now being tested in areas such as communication, report, document review, and decision support.

In AEC-related work, information management is one of the areas where AI can provide value. Construction and engineering projects usually include contracts, drawings, specifications, addendums, schedules, reports, RFIs, and other documents that must be reviewed carefully. Abioye et al. (2021) found that AI has been applied in construction for activities such as risk management, activity monitoring, resource optimization, and project decision-making. These applications show that AI can support companies by helping process large amounts of project information and identifying issues that may affect cost, schedule or performance.

Generative AI is also becoming more relevant in construction because many project tasks involve text-based information. Alwashah et al. (2025) discuss how generative AI can be applied in construction use cases and future workflows. For example, generative AI can help summarize project documents, support communication, organize technical information, and assist with first drafts of reports. This is useful in AEC because many project decisions depend on understanding written requirements and communicating them clearly. However, the literature also suggests that

these tools should be used carefully because AI-generated summaries may not always capture the full context of a project.

A major topic in the literature is that AI adoption depends on more than just access to technology. Regona et al. (2022) found that construction companies face adoption challenges related to cost, training, data quality, trust, and organizational readiness. These challenges are important because companies may see the value of AI but still struggle to implement it correctly. For specialized contractors, this issue can even be stronger because general AI tools may not understand field conditions, specific scopes, or the exact details of each particular project. Because of this, AI adoption in AEC requires both technology and experienced human review.

The literature on financial AI adoption also shows similar concerns. The European Central Bank (2024) explains that AI can support financial work by improving efficiency and helping organizations analyze information, but it can also create risks if not managed properly. Financial work depends on sensitive data, accurate numbers, and professional responsibility. Because of this, AI tools must be used with controls that protect confidential information and reduce the risk of incorrect outputs.

The U.S. Department of Treasury (2024) also discusses the risk associated with AI in financial services, including data privacy, model risk, bias, and dependence on third party providers. These issues are important for consulting companies because consultants often work with private client information. If employees enter sensitive data into AI tools without clear rules, the company could create confidentiality or data protection problems. This shows that AI adoption is not only a productivity issue, but also a governance issue.

The Financial Stability Board (2024) adds that AI use in financial services create concerns related to operational dependency, cyber risk, data gaps, and model governance. These

concerns show that companies need policies and review procedures before relying heavily on AI tools. Even when AI produces useful summaries or analysis, professionals still need to verify the information and understand the limits of the tool.

Overall, the literature shows that AI can improve efficiency and support decision-making, but successful adoption depends on a variety of factors. This connects directly to the companies studied in this report because both construction estimating and financial consulting involve sensitive information and decisions that require accuracy. AI can be valuable, but the literature supports the idea that it should be used as a support tool rather than a replacement for professional judgment.

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